MIL-I-10903(CmiC)
5 Bebruary 1951
SUPERSEDING
Army 96-131-108
22 June 1945

MILITARY SPECIFICATION

ISOBUTYL METHACRYLATE POLYMER, AE

1. SCOPE

1.1 This specification covers one type of powdered isobutyl methacrylate polymer, for use in the manufacture of incendiary oils.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications. - The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATIONS

RR-S-366 - Sieves; Standard, Testing.
VV-B-231 - Benzol (Benzene), Industrial Grade.

MILITARY SPECIFICATIONS

JAN-A-271 - Acid, Stearic
JAN-I-711 - Incendiary Oils, Consistency of; Test Procedures

U. S. ARMY SPECIFICATIONS

94-40645 - Marking; Exterior, Domestic and Export Shipment, by Contractors
100-2 - Standard Specification for Marking Shipments by Contractors

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.

(Army. - Copies of specifications may be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(Navy. - Copies of Federal, Military, and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y.)

(Air Force. - Copies of Federal, Military, and U. S. Army specifications may be obtained upon application to the Commanding General, Air Development Force, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

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2.2 Other publications.- The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y.

INTERSTATE COMMERCE COMMISSION PUBLICATION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained from the Interstate Commerce Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Particle size.- The material shall conform to the particle size requirements shown in table I.

Table I.- Particle size

Passing	Held On	Maximum	Minimum
No. 10 sieve*	----	-----	100%
-----	No. 20 sieve*	-----	10%
No. 20 sieve*	No. 50 sieve*	75%	35%
No. 50 sieve*	No. 100 sieve*	35%	---
No. 100 sieve*	----	20%	-----

* U. S. Standard sieves

3.2 Solubility.- The material shall form a substantially clear, smooth solution when dissolved in a mixture of solvent blend and stearic acid.

3.3 Gelling power- The material, when treated under standardized conditions, shall form a gel with the properties shown in table II.

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Table II. - Gelling power

Property	Maximum	Minimum
Elasticity	-----	320°
Consistency	3.2 minutes	1.0 minutes

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. - A lot shall consist of the isobutyl methacrylate polymer AE produced by one manufacturer with no change in process or materials in no more than 24 consecutive hours, providing the operation is continuous. In the event the process is a batch operations, each batch shall constitute a lot (see 6.2).

4.2 Sampling. - Ten specimens shall be selected as representative of each lot. The specimens may be taken from ten randomly selected containers or at intervals throughout the container filling operation. The specimens shall be placed in clean, dry containers, and labeled to identify the container represented.

4.3 Inspection.

4.3.1 Naval purchases. - For Naval purchases, the general inspection procedures shall be in accordance with the General Specifications for Inspection of Material.

4.3.2 Packing and marking. - The inspector shall inspect the packing and marking of the material for conformance with section 5.

4.4 Tests. - All specimens shall be examined for compliance with the solubility requirement. Inspection may be stopped if any specimen fails to meet this requirement. Providing all specimens meet the solubility requirement, three of the ten specimens shall be subjected to the tests of 4.4.2, 4.4.3.2, and 4.4.3.3.

4.4.1 Reagents. - Reagents shall conform to the requirements shown in table III. Distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with each test, using the same quantities of reagents used in the test, and corrections shall be applied when significant.

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Table III. - Reagent requirements

Reagent	Requirements	Maximum	Minimum
Cyclohexane (Commercial grade)	Refractive index (D line, 20°C.)	1.4270	1.4240
	Melting point	6.0°C	4.0°C
Benzene (Specification VV-B-231)	*Refractive index (D line, 20°C.)	1.5040	1.4990
	*Melting point	5.5°C	5.2°C
Solvent blend (95 parts cyclohexane, 5 parts benzene by weight)	Refractive index (D line, 20°C.)	1.4285	1.4277
	Melting point	-5.0°C	-8.0°C
Stearic acid (Specification JAN-A-271)	* Acid number	209	205
	*Iodine number	4.0	----
Calcium hydroxide (reagent grade)	$Ca(OH)_2$ content	----	95%
	Iron	Trace	----
	Magnesium	Trace	----

* Requirement not included in specification

4.4.2 Particle size. -Assemble a nest of U. S. Standard sieves conforming to Specification RR-S-366, consisting of a No. 10, No. 20, No. 50, and No. 100 sieve, with the largest mesh on top and a receiving pan at the bottom. Place 100.0 gm. of specimen in the No. 10 sieve, cover, and assemble the nest of sieves in a mechanical shaker geared to produce 300 ± 10 taps of the striker per minute. Operate the shaker for 5 minutes. Weigh the portions retained or passed by the various sieves and calculate the results to a percent of the total weight recovered.

4.4.3 Solubility and gelling power.

4.4.3.1 Solubility test. - Weigh 247 gm. of solvent blend (see table III), at 25.0° ± 0.5°C., ;into a tared, square-type pint mason jar. Add 9.0 gm. of powdered stearic acid (see table III) and stir with a mechanical stirrer rotating at a speed of 500 ± 10 rpm. After 1 minute add 22.5 gm. of the polymer specimen, cover the jar loosely and continue to stir for 30 minutes. Maintain the temperature at 25.0° ± 0.5°C. throughout the stirring period. At the end of the stirring period examine the solution visually for conformance to the requirements of 3.2. Use the solution in the gelling power test (see 4.4.3.2).

4.

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4.4.3.2 Preparation of gel. - Within 10 minutes after completing the test specified in 4.4.3.1 add to the solution a smooth slurry, containing 6.0 gm. of calcium hydroxide (100 percent through a No. 100 sieve) in 12.0 gm. of solvent blend, and stir for 1 minute with a mechanical stirrer rotating at a speed of 500 ± 10 rpm. The stirrer shall be made of stainless steel or glass, and shall have the following dimensions: the rod shall be 3/8 inch in diameter by 8-1/4 inches long and fitted with two vanes 1 by 1/2-by 1.8-inch, set at right angles and with the faces of the vanes at a 15-degree angle to the axis of the rod. Maintain the temperature of the solution at 25.0° ± 5°C. during the test. After the 1-minute stirring period add 3.75 ml. of water and stir an additional 5 minutes, moving the jar up and down on the stirring rod to assure thorough mixing. After this mixing period, seal the jar tightly and store for 24.0 ± 5 hours at 25.0° ± 0.5°C.

4.4.3.3 Gel tests. - After storage, the prepared gel shall be tested for elasticity and consistency by the torsiometer method described in Specification JAN-I-711 with the following exceptions:

a. A 20-gm. weight shall be used for the consistency test instead of the 70-gm. weight normally supplied. The 500-gm. weight is used for elasticity tests.
b. The temperature of the sample shall be maintained at 25° ± 1°C. at the time of testing; record the temperature with the test results.
c. Because of the small forces involved special care shall be exercised to insure that the pointer is accurately set on the zero position, with no stress on the paddle or gel, before each test.

4.5 Rejection and resubmission. - If any specimen fails to conform to this specification, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made of each container in the lot and removing the nonconforming containers at no expense to the Government. The contractor may then resubmit the remaining portion of the lot for acceptance testing.

5. PREPARATION FOR DELIVERY

5.1 Packing. - The material shall be packed in 100-pound capacity fiber drums, conforming to Specification 21A of the Interstate Commerce Commission Regulations for transportation of Explosives and Other Dangerous Articles, etc. Each drum shall be fitted with a polyethylene bag constructed of polyethylene sheet having a nominal thickness of 0.004 inch. After filling, the bag shall be closed by knotting, tying, or heat sealing.

5.2 Marking. - In addition to any special marking required by the contract or order, all shipping containers for the Army shall be marked in accordance with Specification 100-2; for the Air Force, in accordance with Specification 94-40645; and for the Navy, in accordance with the Navy Shipment Marking Handbook.

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number, and date of this specification.

6.2 Batch. - A batch is defined as that quantity of material which has been subjected to some chemical or physical mixing process intended to make the final product substantially uniform.

6.3 Inspection. - Inspection will not necessarily include all drawing and specification requirements; however, the omission of inspection does not constitute a waiver of these requirements.

6.4 Sampling and testing. - If the contractor consistently produces high-quality material and operates under a system of quality con rol acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

Custodian:

Army - Chemical Corps.

Other interest:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426
1 APR 63

REPLACES NAVSHIPS FORM 4003, WHICH IS OBSOLETE

PLATE NO. 15419

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